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TEC–801

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

SATELLITE COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain Kepler's law. What are some examples of how Kepler's laws can be used to predict the behaviour of satellites in orbit ? (CO1)

OR

(b) Write short notes on LEO, MEO and GEO orbits. (CO1)

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2. (a) What are the key properties of a satellite communication system ? (CO1)

OR

(b) What are orbital perturbations ? What are some strategies that can be used to mitigate the effects of orbital perturbations ? (CO1)

3. (a) What are the different types of Earth stations ? How do Earth stations communicate with satellites in orbit ? (CO2)

OR

(b) What is orbit control system of a satellite and what is its purpose ? (CO2)

4. (a) How do Telemetry, Tracking, Command and Monitoring (TTC and M) systems work and what role do they play in satellite operations ? (CO2)

OR

(b) What are the key components of a communication subsystem on a satellite

and how do they work together to enable communications with Earth stations ? (CO2)

5. (a) What is the role of the antenna subsystem in a satellite and what factors determine the choice of antenna design ? (CO2)

OR

(b) How does a satellite's power system work and what are some of the challenges of powering a spacecraft in orbit ? (CO2)